

Machine Learning

CMPSCI 589

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Evaluation Metrics (1/2)

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College of Information
& Computer Sciences



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Review:

Document Classification

Multinomial Naive Bayes

Review: Multinomial Naive Bayes Classification

Multinomial model: each entry indicates the frequency of a word in the document

$\mathbf{m} = (2, 1, 0, 1, 0, 1, 0)$

	password	expired	send	review	conference	account	paper
doc1	2	1	0	1	0	1	0

- To classify a new document, compute the following for each class y_i :

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Review: Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- $\text{len}(\text{Doc})$ is the length of Doc (i.e., the number of unique words that appear in that document)
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the frequency of the word w_k in documents belonging to class y_i

$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

total number of times word w_k
appears in documents of class y_i
divided by
total number of words that
appear in documents of class y_i



the relative frequency of word w_k in documents of class y_i

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- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the frequency of the word w_k in documents belonging to class y_i

In practice, only words that appear in the document being classified need to be taken into account

percentage of
documents
of class y_i

total number of times word w_k
appears in documents of class y_i
divided by

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the relative frequency of word w_k in documents of class y_i

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$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
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w_8	6	6/36	1	1/16

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w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 6/11 \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= 6/11 \times$$

$$2/36 \times$$

$$4/36 \times$$

$$1/36 \times$$

$$7/36$$

$$= 0.0000181$$

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$$\begin{aligned} \Pr(y = \text{SP} \mid \text{Doc}) &= \frac{6}{11} \times \Pr(w_3 \mid \text{SP}) \times \Pr(w_5 \mid \text{SP}) \times \Pr(w_2 \mid \text{SP}) \times \Pr(w_7 \mid \text{SP}) \\ &= \frac{6}{11} \times \frac{2}{36} \times \frac{4}{36} \times \frac{1}{36} \times \frac{7}{36} \\ &= 0.0000181 \end{aligned}$$

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 \Pr(y = \text{SP} \mid \text{Doc}) &= \frac{6}{11} \times \Pr(w_3 \mid \text{SP}) \times \Pr(w_5 \mid \text{SP}) \times \Pr(w_2 \mid \text{SP}) \times \Pr(w_7 \mid \text{SP}) \\
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$$\begin{aligned}
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$$\begin{aligned}
 \Pr(y = \text{SP} \mid \text{Doc}) &= \frac{6}{11} \times \\
 &\quad \Pr(w_3 \mid \text{SP}) \times \\
 &\quad \Pr(w_5 \mid \text{SP}) \times \\
 &\quad \Pr(w_2 \mid \text{SP}) \times \\
 &\quad \Pr(w_7 \mid \text{SP}) \\
 &= \frac{6}{11} \times \\
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w_6	6	6/36	2	2/16
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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = 5/11 \times$$

$$\Pr(w_3 \mid \text{INF}) \times$$

$$\Pr(w_5 \mid \text{INF}) \times$$

$$\Pr(w_2 \mid \text{INF}) \times$$

$$\Pr(w_7 \mid \text{INF})$$

$$= 5/11 \times$$

$$3/16 \times$$

$$1/16 \times$$

$$4/16 \times$$

$$3/16$$

$$= 0.000249$$

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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = \frac{5}{11} \times$$

$$\Pr(w_3 \mid \text{INF}) \times$$

$$\Pr(w_5 \mid \text{INF}) \times$$

$$\Pr(w_2 \mid \text{INF}) \times$$

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$$= \frac{5}{11} \times$$

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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

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$$\Pr(w_3 \mid \text{INF}) \times$$

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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = \frac{5}{11} \times$$

$$\Pr(w_3 \mid \text{INF}) \times$$

$$\Pr(w_5 \mid \text{INF}) \times$$

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$$\Pr(w_7 \mid \text{INF})$$

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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

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$$\Pr(w_3 \mid \text{INF}) \times$$

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	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \text{Doc}) &= \frac{5}{11} \times \\ &\quad \Pr(w_3 \mid \text{INF}) \times \\ &\quad \Pr(w_5 \mid \text{INF}) \times \\ &\quad \Pr(w_2 \mid \text{INF}) \times \\ &\quad \Pr(w_7 \mid \text{INF}) \\ &= \frac{5}{11} \times \\ &\quad \frac{3}{16} \times \\ &\quad \frac{1}{16} \times \\ &\quad \frac{4}{16} \times \\ &\quad \frac{3}{16} \end{aligned}$$

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$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

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The new document seems to belong to the “Informatics” class

Data Pre-Processing

- **Pre-processing the text** in the documents is important
 - *Stemming*: removes the suffix of words so they are treated as being equal
 - Fishing, fisher, fished → fish
 - Removing "*stop words*" → common words such as “the”, “of”, etc.

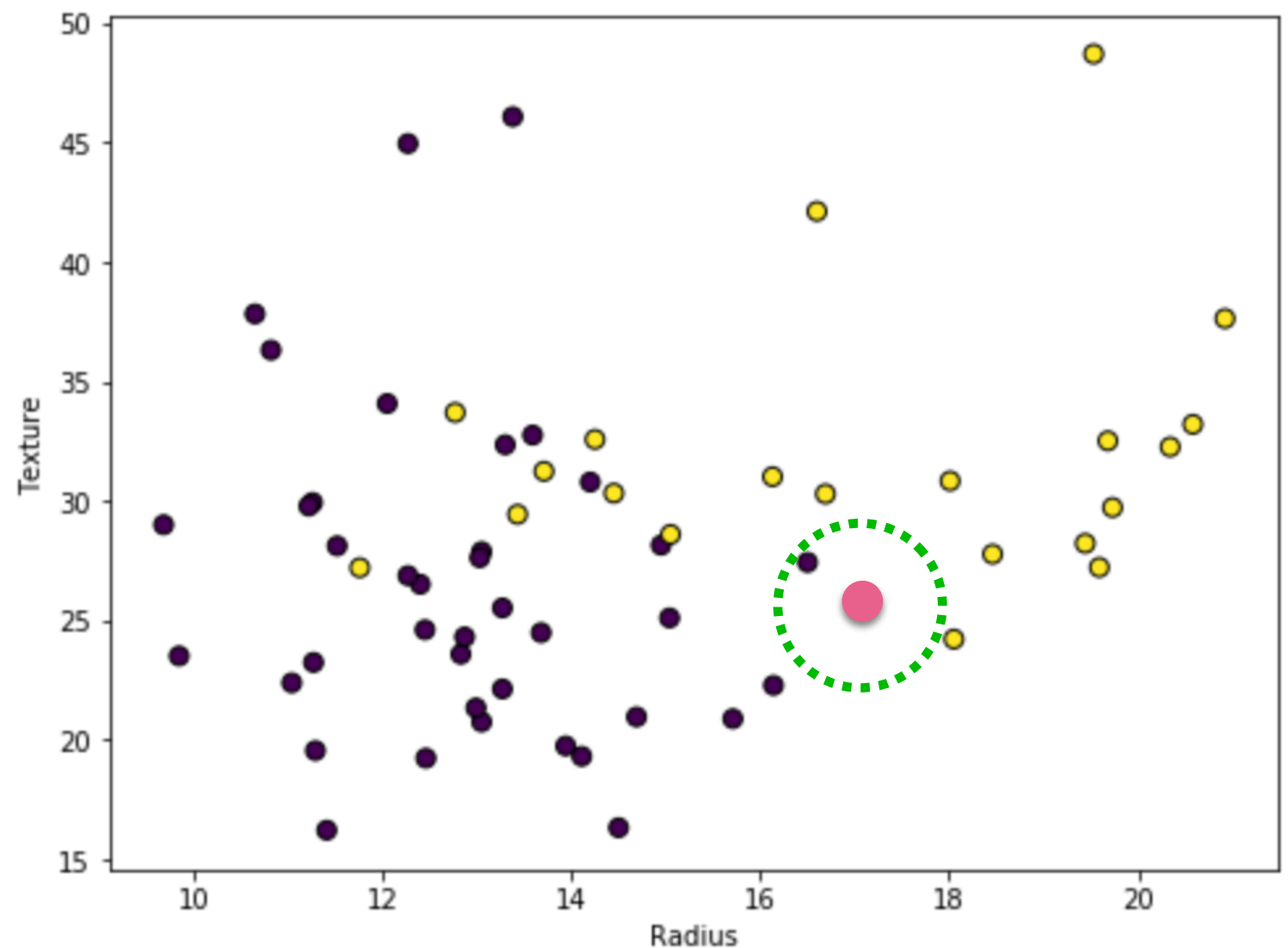
How to evaluate if a trained ML model is “good”?

(and how to choose its hyperparameters so that it works well?)

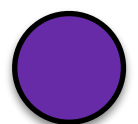
Revisiting k -NN

- To classify a **new instance** (with unknown class/label) its **k nearest instances** vote

- parameter of the algorithm
- selected by the user
- depends on the application



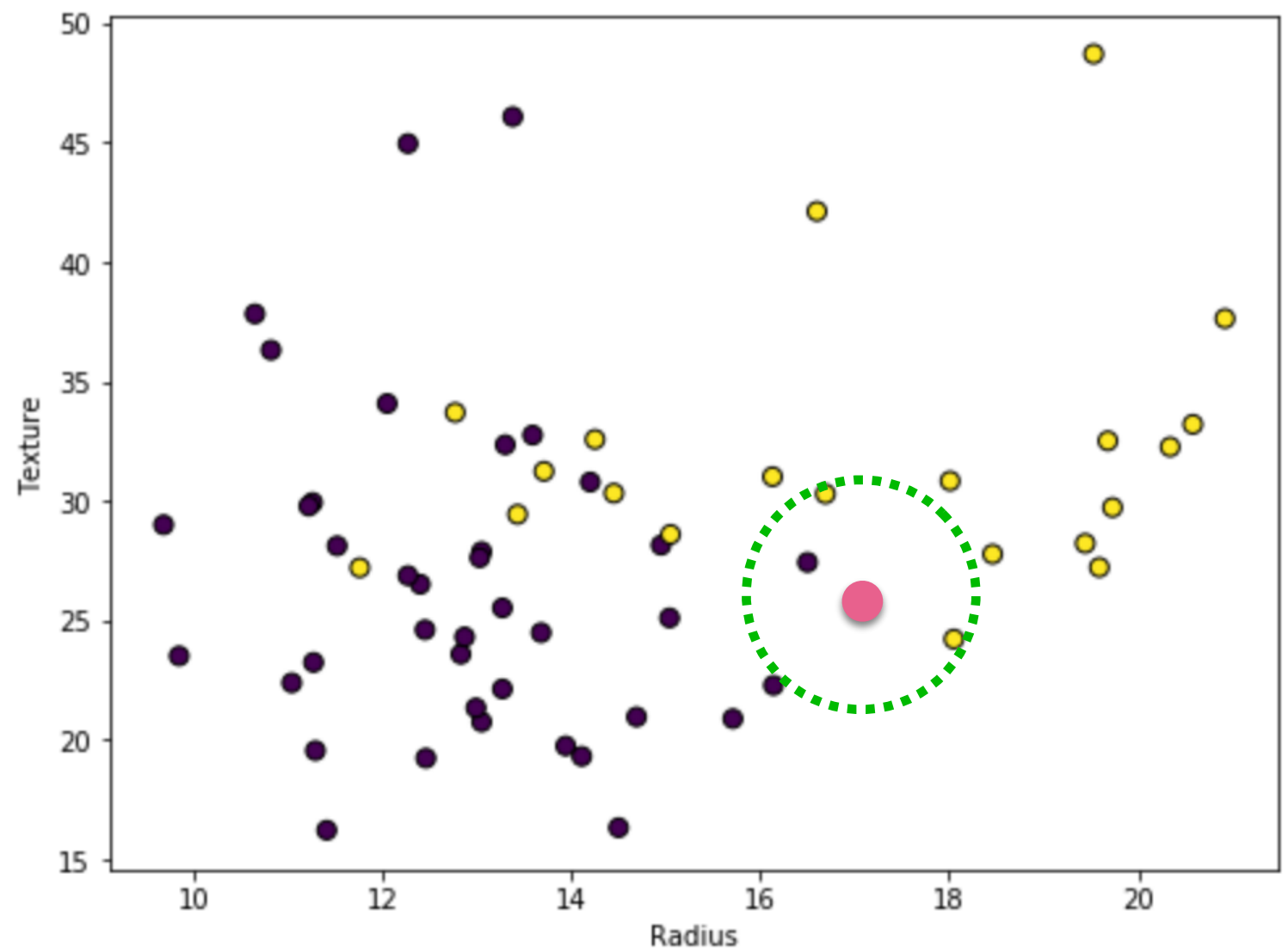
Assume $k=1$
What is the predicted class?



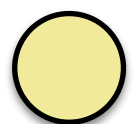
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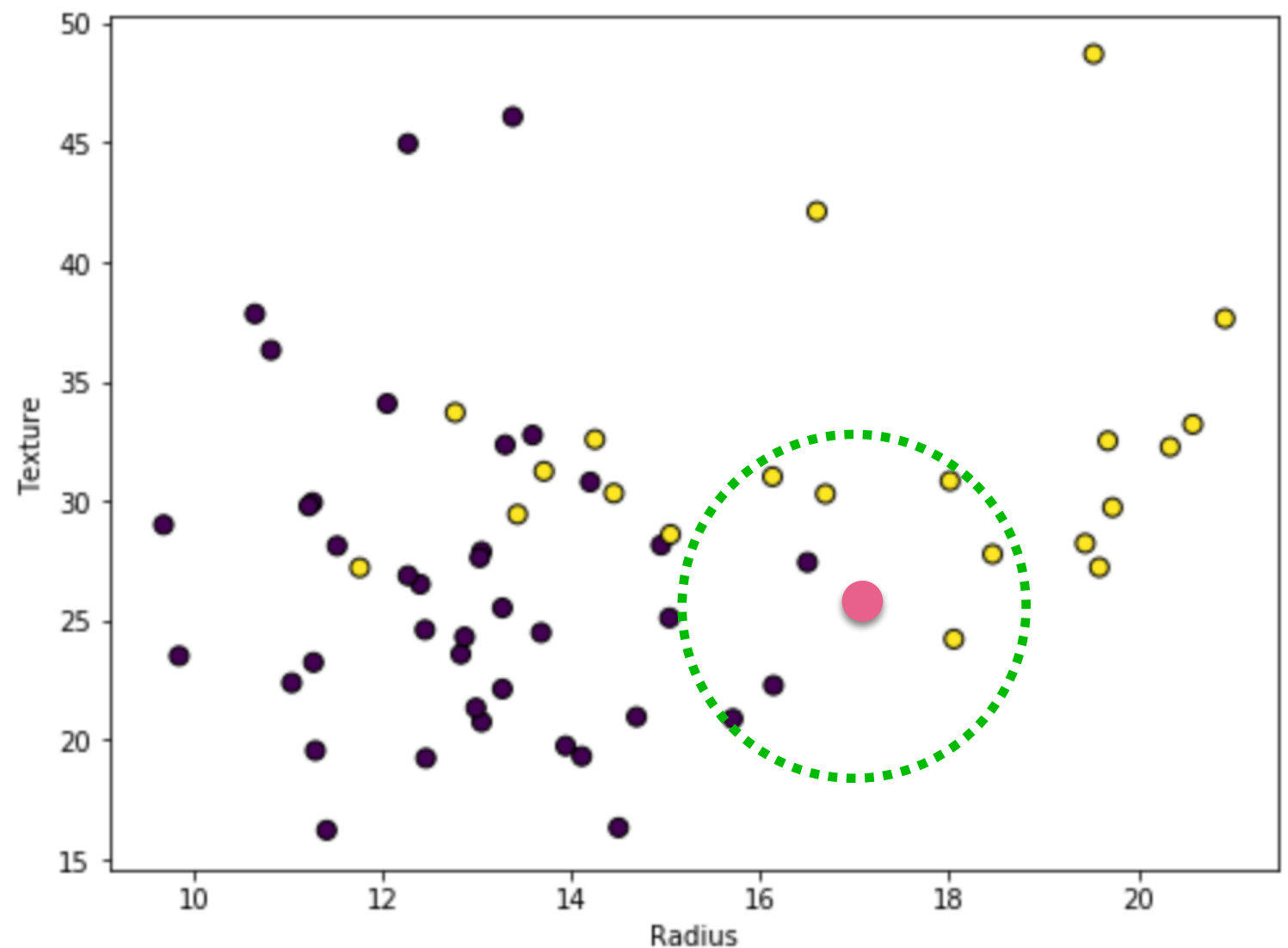
Assume $k=3$
What is the predicted class?



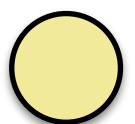
Revisiting k -NN

- To classify a new instance (with unknown class/label) its k nearest instances vote

- parameter of the algorithm
- selected by the user
- depends on the application

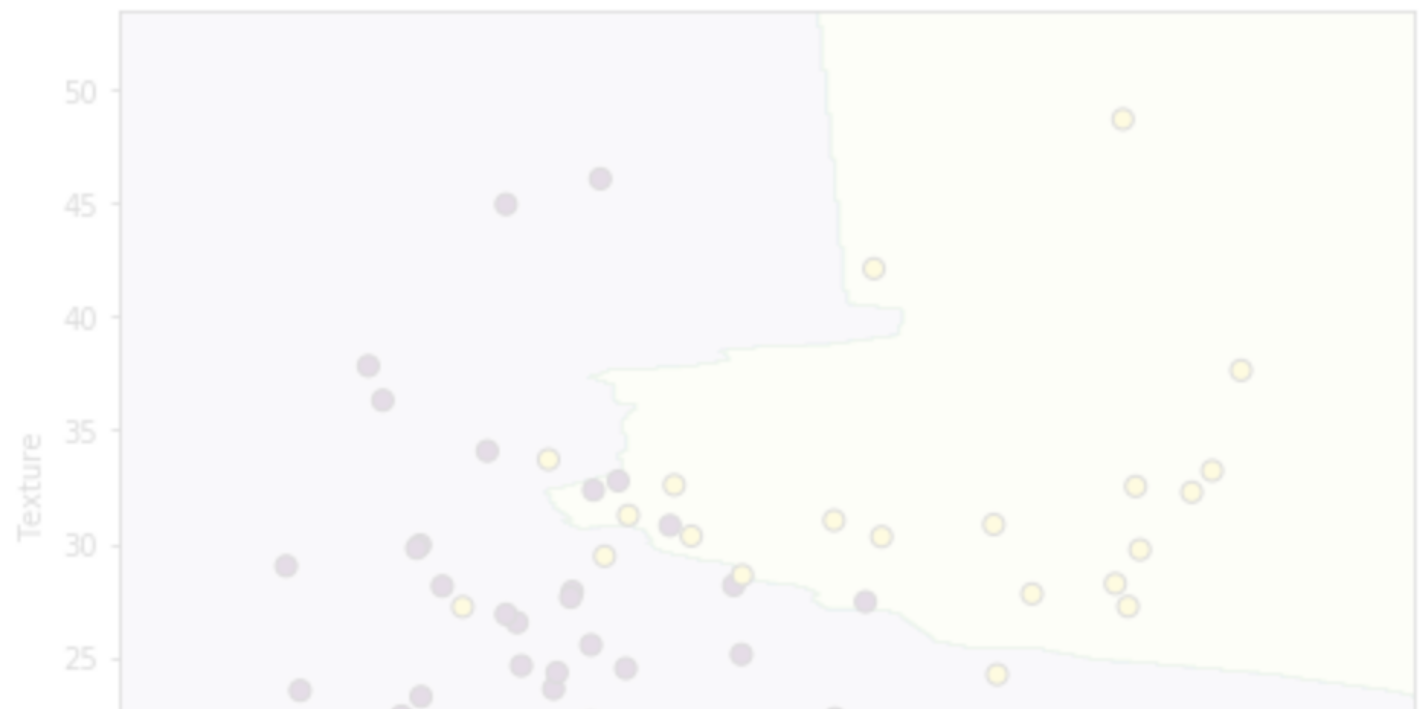


Assume $k=8$
What is the predicted class?



Revisiting k -NN

$k=13$



- To classify a new instance
(with unknown class/label)

Which value of k results in the best model?

Although extremely simple, the k -NN algorithm
is capable of learning complex non-linear **decision boundaries**

How to determine what is the best model?

Multiple ML algorithms may be used

- 1) We need to select which ML algorithm to use
- 2) Given that we're using that algorithm, we may also have to tune its hyperparameters

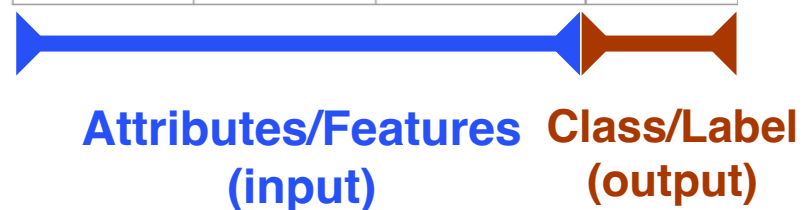
- Needs to measure/validate the performance of different types of models
- Select the best one

Generalization

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
9	21	55	B
9	14	59	B
9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B



Recall:

- Goal is *not* to “learn” just about the examples in the dataset
- Instead, we want to **generalize such knowledge**
- In order to **correctly predict the label of new/unknown instances**

17	23	98	??
----	----	----	----

New (unknown/unlabeled) patient

Generalization

Generalization is the main goal in ML

Intuitively:

- Learn a model based on training data...
- Which can be later on used to make “good” predictions about new, unknown data
- How to quantify a model’s capability of generalizing?

15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
11	11	55	B
9	14	59	B
9	21	60	B
11	15	72	B
13	16	81	B

Attributes/Features (input) Class/Label (output)

17	23	98	??
----	----	----	----

New (unknown/unlabeled) patient

• instead, we want to generalize such knowledge

• in order to correctly predict the label of new/unknown instances

Generalization

Generalization is the main goal in ML

Evaluating a model using the same data used to train it → bad idea!

Overestimates performance because model may have "memorized" training data (overfitting)

17

23

98

??

New (unknown/unlabeled) patient

Holdout Method

Splits the existing examples into training set and a test set

	Attributes	Class/Label
Instance1	1.17	A
Instance2	2.97	B
Instance3	3.26	B
Instance4	4.69	B
...		
Instance600	6.00	B
Instance601	6.41	A
...		
Instance700	6.00	A
Instance701	6.41	A
...		
Instance999	3.26	A
Instance1000	4.69	B

**Training
set
(70%)**

**Test
set
(30%)**

**Completely
disjoint!**

Holdout Method

	Attributes	Class/Label
Instance1	Training set	
Instance2		
Instance3		
Instance4		
...		
Instance600		
Instance601		
...		
Instance700		
Instance701		
...	Test set	
Instance999		
Instance1000		

Splits the existing examples into training set and a test set

Training set (70%)

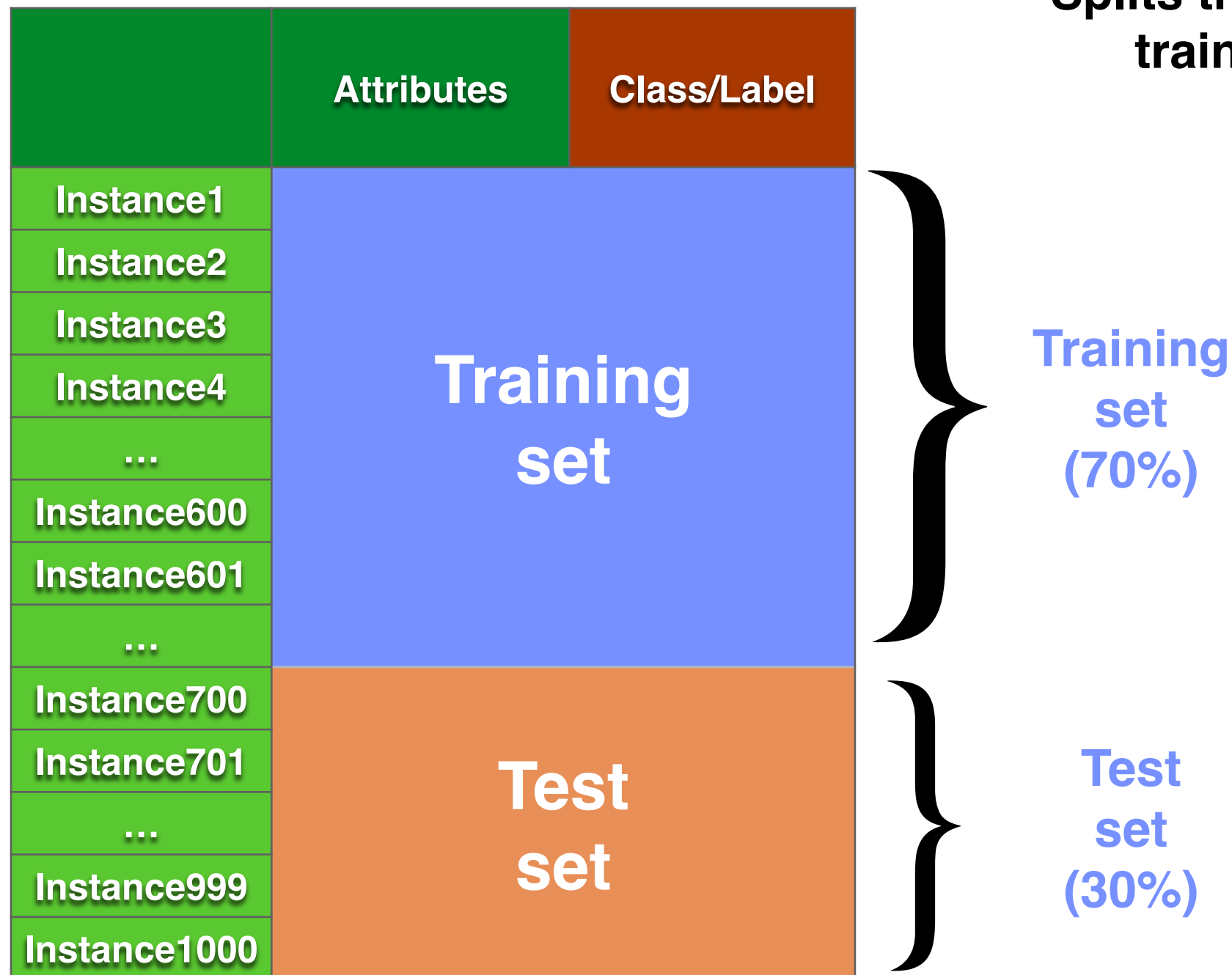
Used to train the machine learning model

Test set (30%)

Used to quantify how well the learned model generalizes

Holdout Method

Splits the existing examples into training set and a test set



Sometimes:

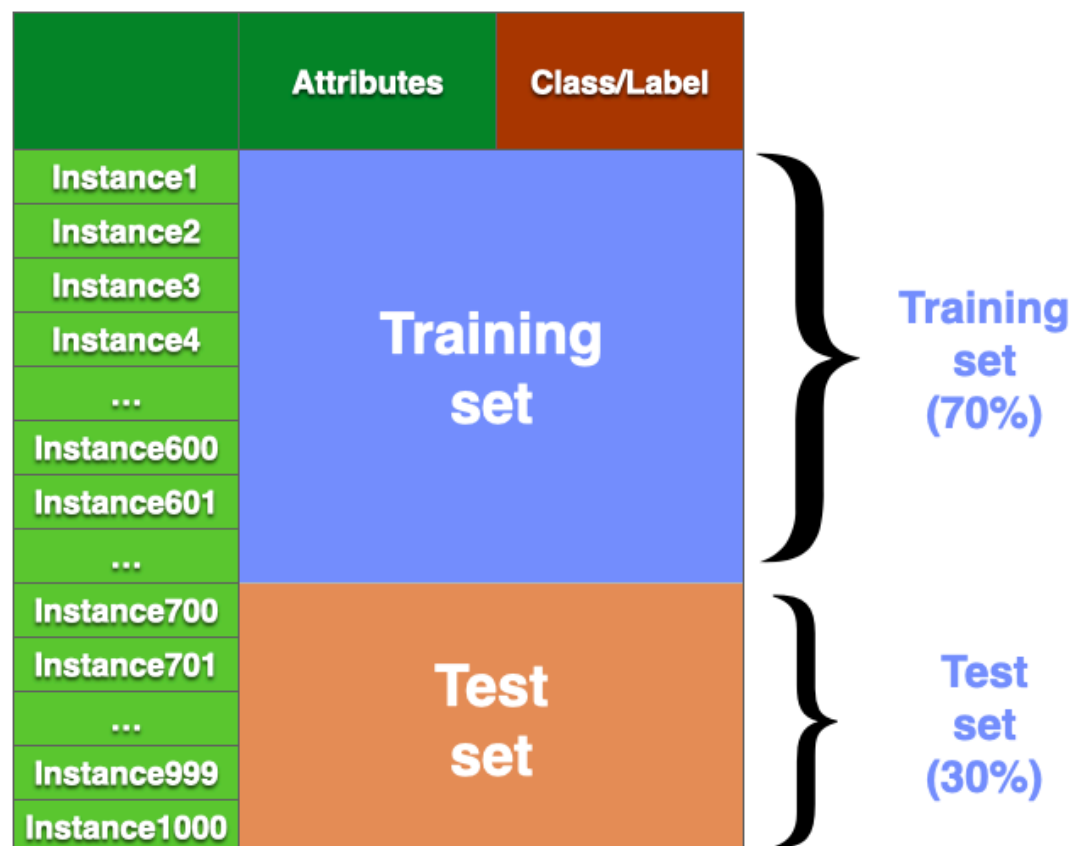
90% vs 10%

80% vs 20%

60% vs 40%

...

Holdout Method



1. Shuffles the dataset and randomly splits it into training and test (70% vs 30%)

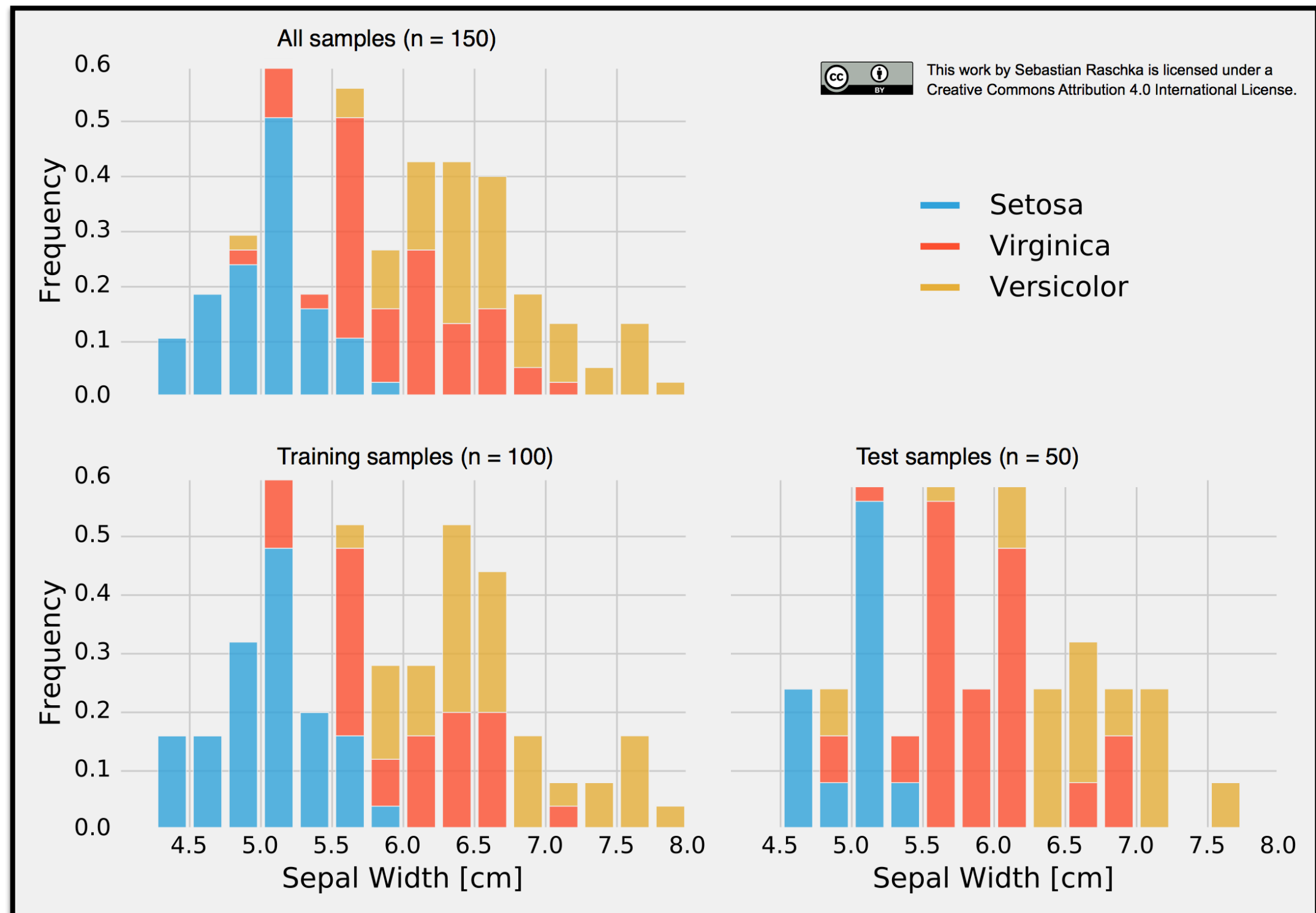
2. Keeps the same proportions of classes both in the training and test sets
(stratified holdout)

Iris dataset:
33.3% Setosa
33.3% Virginica
33.3% Versicolor

random split,
not stratified

Training dataset
38% Setosa
28% Virginica
34% Versicolor

Test dataset
24% Setosa
44% Virginica
32% Versicolor



Iris dataset:

33.3% Setosa
33.3% Virginica
33.3% Versicolor

random split,
not stratified



Training dataset

38% Setosa
28% Virginica
34% Versicolor

Test dataset

24% Setosa
44% Virginica
32% Versicolor

- **Training and test sets do not have the same class distribution as the original dataset**
- **Might cause poor estimates of generalization when evaluating the model on the test set**
(this happens especially if the training set is small)
- **When splitting into training and test datasets**
 - we should **perform stratified holdout**
 - i.e., **preserve the class distribution of the original dataset both on the training and test datasets**

Iris dataset:
33.3% Setosa
33.3% Virginica
33.3% Versicolor

random split,
not stratified



Training dataset
38% Setosa
28% Virginica
34% Versicolor

Test dataset
24% Setosa
44% Virginica
32% Versicolor

Example

Suppose the dataset has **1000** instances

Three classes: **C1, C2, C3**

Proportion of instances of each class:

C1: 50% C2: 30% C3: 20%

Assume we will split the dataset as follows:

800 training // **200** test

Iris dataset:
33.3% Setosa
33.3% Virginica
33.3% Versicolor

random split,
not stratified



Training dataset
38% Setosa
28% Virginica
34% Versicolor

Test dataset
24% Setosa
44% Virginica
32% Versicolor

Example

Suppose the dataset has **1000** instances

Proportion of instances of each class:

C1: 50% C2: 30% C3: 20%

Assume we will split the dataset as follows:

800 training // 200 test

Determining number of instances of each class (**stratified training**):

50% of 800 = **400**

30% of 800 = **240**

20% of 800 = **160**

Determining number of instances of each class (**stratified testing**):

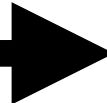
50% of 200 = **100**

30% of 200 = **60**

20% of 200 = **40**

Iris dataset:
33.3% Setosa
33.3% Virginica
33.3% Versicolor

random split,
not stratified



Training dataset
38% Setosa
28% Virginica
34% Versicolor

Test dataset
24% Setosa
44% Virginica
32% Versicolor

Example

Suppose the dataset has **1000** instances

Proportion of instances of each class:

C1: 50% C2: 30% C3: 20%

Determining number of instances of each class (**stratified training**):

50% of 800 = **400**

30% of 800 = **240**

20% of 800 = **160**

Determining number of instances of each class (**stratified testing**):

50% of 200 = **100**

30% of 200 = **60**

20% of 200 = **40**

Let's check:

C1: 400 + 100 = 500 (50% of 1000)

C2: 240 + 60 = 300 (30% of 1000)

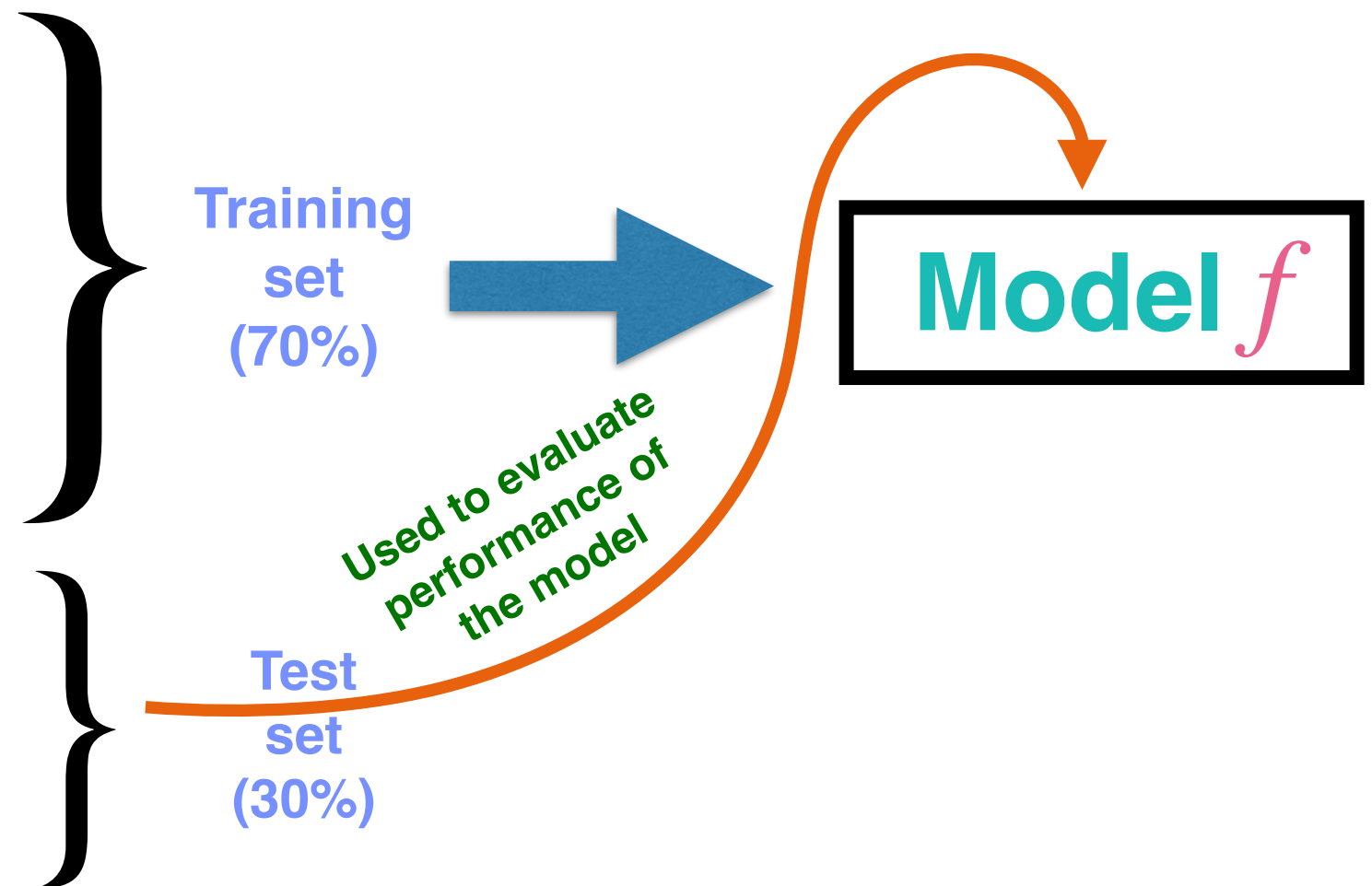
C3: 160 + 40 = 200 (20% of 1000)



Holdout Method

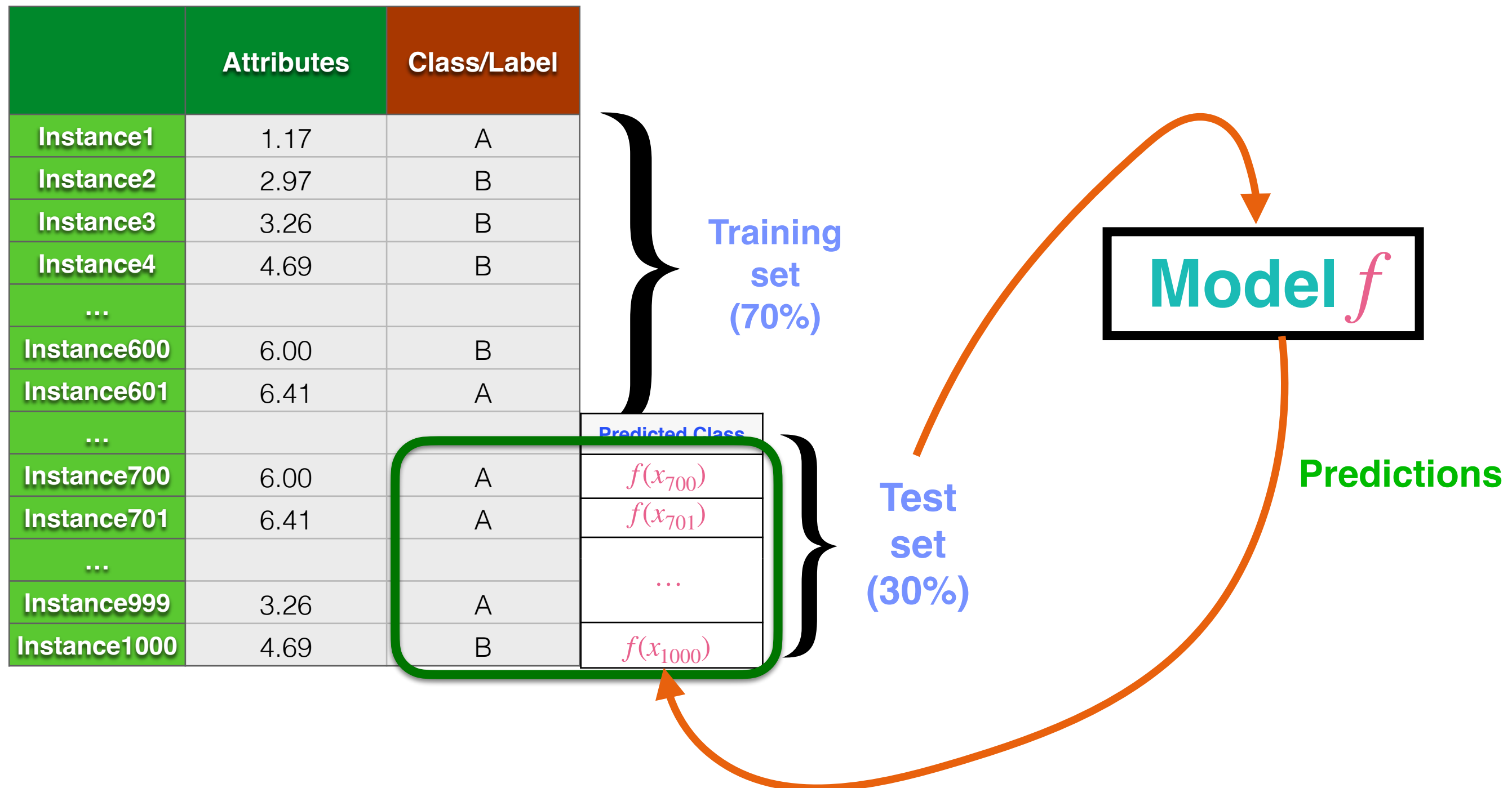
- Splits dataset into training and test sets
(possibly via stratified partitioning)

	Attributes	Class/Label
Instance1	1.17	A
Instance2	2.97	B
Instance3	3.26	B
Instance4	4.69	B
...		
Instance600	6.00	B
Instance601	6.41	A
...		
Instance700	6.00	A
Instance701	6.41	A
...		
Instance999	3.26	A
Instance1000	4.69	B



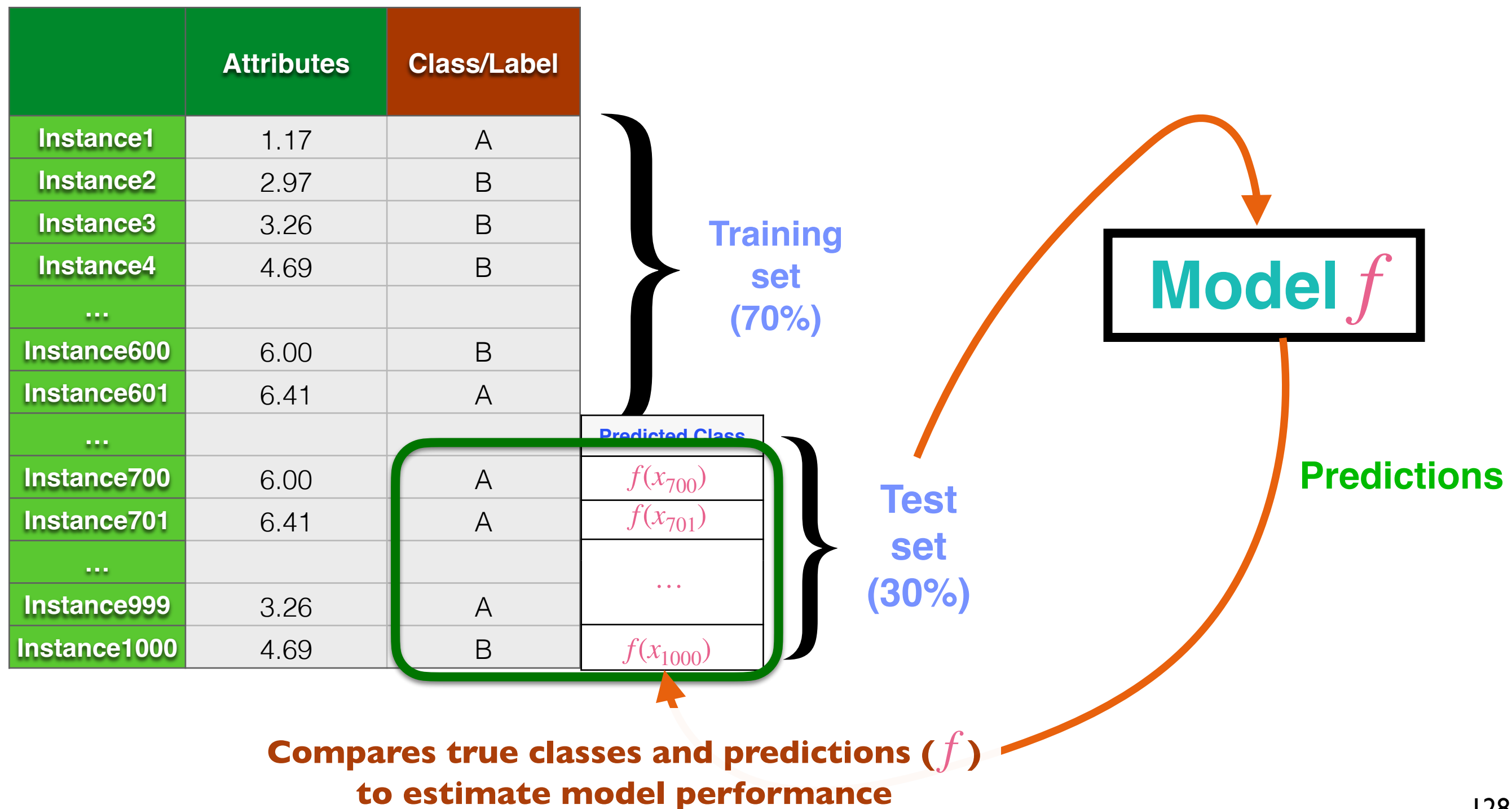
Holdout Method

- Splits dataset into training and test sets
(possibly via stratified partitioning)



Holdout Method

- Splits dataset into training and test sets
(possibly via stratified partitioning)



Holdout Method

- Splits dataset into training and test sets
(possibly via stratified partitioning)

	Attributes	Class/Label	
Instance1	1.17	A	Training set (70%)
Instance2	2.97	B	
Instance3	3.26	B	
Instance4	4.69	B	
...			
Instance600	6.00	B	
Instance601	6.41	A	
...			Test set (30%)
Instance700	6.00	A	
Instance701	6.41	A	
...			
Instance999	3.26	A	
Instance1000	4.69	B	

Model f

performance = x

- Repeats this process many times
- (i.e., splitting dataset, evaluating on test set)
- Each time, training a different model
 - k-NN
 - decision tree
 - neural network
 - etc.

Holdout Method

- Splits dataset into training and test sets
(possibly via stratified partitioning)

	Attributes	Class/Label
Instance1	1.17	A
Instance2	2.97	B
Instance3	3.26	B
Instance4	4.69	B
...		
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Instance601	6.41	A
...		
Instance700	6.00	A
Instance701	6.41	A
...		
Instance999	3.26	A
Instance1000	4.69	B

Training
set
(70%)

Test
set
(30%)

k -NN

Model f_1

performance = x

Decision Tree

Model f_2

performance = y

...

Neural Network

Model f_k

performance = z

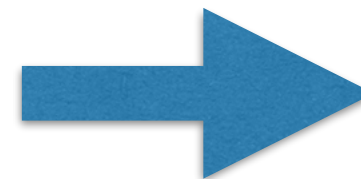
Selects the model with highest performance

Holdout Method

- Splits dataset into training and test sets
(possibly via stratified partitioning)

	Attributes	Class/Label
Instance1	1.17	A
Instance2	2.97	B
Instance3	3.26	B
Instance4	4.69	B
...		
Instance600	6.00	B
Instance601	6.41	A
...		
Instance700	6.00	A
Instance701	6.41	A
...		
Instance999	3.26	A
Instance1000	4.69	B

Entire
dataset



Decision Tree

Model f_2

Trains the selected model
with *all* available data

→ Deploy the final trained model



Selects the model with highest performance

Performance Metrics



Performance is measured by comparing predictions with the true class/output

- There are different types of metrics...
 - depending on whether it is a [classification](#) or [regression](#) problem

Simplest metrics to quantify the performance of classifiers

Error Rate and Accuracy

Classification Performance - Error Rate

- **Error Rate**

- Percentage of instances **incorrectly** classified
- It is a value between 0 and 1
 - 0 = all predictions are correct
 - 1 = all predictions are incorrect

Model f

$$\text{err}(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{I} (y_i \neq f(\mathbf{x}_i))$$

where $\mathbf{I}(a)$ is the *indicator function*:

$$\mathbf{I}(\mathbf{a}) = \begin{cases} 1, & \text{if } a \text{ is true} \\ 0, & \text{otherwise} \end{cases}$$

Classification Performance - Error Rate

- **Error Rate**

- Percentage of instances **incorrectly** classified
- It is a value between 0 and 1
 - 0 = all predictions are correct
 - 1 = all predictions are incorrect

$$\mathbf{err}(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{I} \left(y_i \neq f(\mathbf{x}_i) \right)$$

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$$\mathbf{I}(\mathbf{a}) = \begin{cases} 1, & \text{if } a \text{ is true} \\ 0, & \text{otherwise} \end{cases}$$

	Attributes	Class/ Label	Prediction
Instance1	1.17	A	A
Instance2	2.97	B	A
Instance3	3.26	B	B
Instance4	4.69	B	A
Instance5	3.24	A	B

$$\begin{aligned} \mathbf{err}(f) &= \frac{1}{5} (0 + 1 + 0 + 1 + 1) \\ &= \frac{3}{5} = 60 \% \end{aligned}$$

Classification Performance - Accuracy

- **Accuracy**

- Percentage of instances **correctly** classified
- It is a value between 0 and 1
 - 0 = all predictions are incorrect
 - 1 = all predictions are correct

Model f

$$\mathbf{acc}(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{I} \left(y_i = f(\mathbf{x}_i) \right)$$

where $\mathbf{I}(a)$ is the *indicator function*:

$$\mathbf{I}(\mathbf{a}) = \begin{cases} 1, & \text{if } a \text{ is true} \\ 0, & \text{otherwise} \end{cases}$$

Classification Performance - Accuracy

- **Accuracy**

- Percentage of instances correctly classified
- It is a value between 0 and 1
 - 0 = all predictions are incorrect
 - 1 = all predictions are correct

$$\text{acc}(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{I} (y_i = f(\mathbf{x}_i))$$

where $\mathbf{I}(a)$ is the *indicator function*:

$$\mathbf{I}(\mathbf{a}) = \begin{cases} 1, & \text{if } a \text{ is true} \\ 0, & \text{otherwise} \end{cases}$$

	Attributes	Class/ Label	Prediction
Instance1	1.17	A	A
Instance2	2.97	B	A
Instance3	3.26	B	B
Instance4	4.69	B	A
Instance5	3.24	A	B

$$\begin{aligned} \text{acc}(f) &= \frac{1}{5} (1 + 0 + 1 + 0 + 0) \\ &= \frac{2}{5} = 40\% \end{aligned}$$

Classification Performance - Accuracy

- **Accuracy**

- Percentage of instances correctly classified
- It is a value between 0 and 1
 - 0 = all predictions are incorrect
 - 1 = all predictions are correct

$$\text{err}(f) = 0.6$$

	Attributes	Class/Label	Prediction
Instance1	1.17	A	A
Instance2	2.97	B	A
Instance3	3.26	B	B
Instance4	4.69	B	A
Instance5	3.24	A	B

Model f

$$\begin{aligned}\text{acc}(f) &= 1 - \text{err}(f) \\ &= 1 - 0.6 \\ &= 0.4\end{aligned}$$

Classification Performance

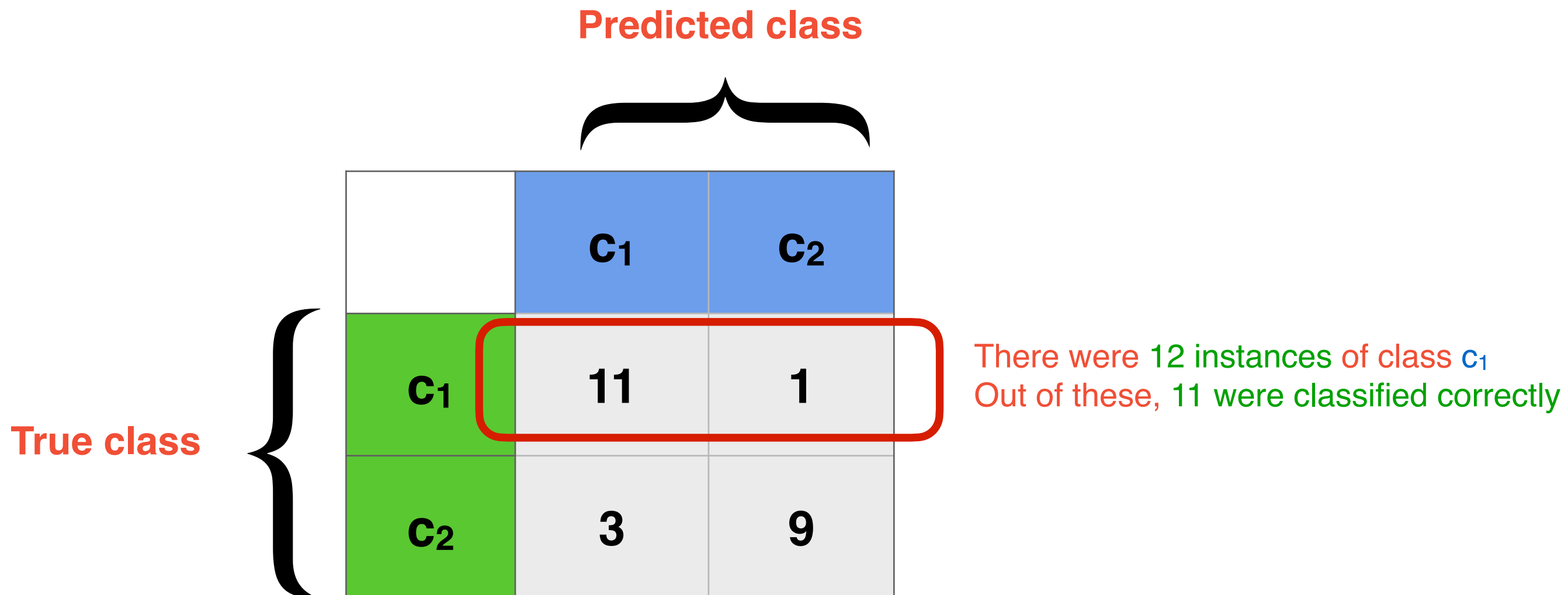
- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)

Predicted class

	c_1	c_2
c_1	11	1
c_2	3	9

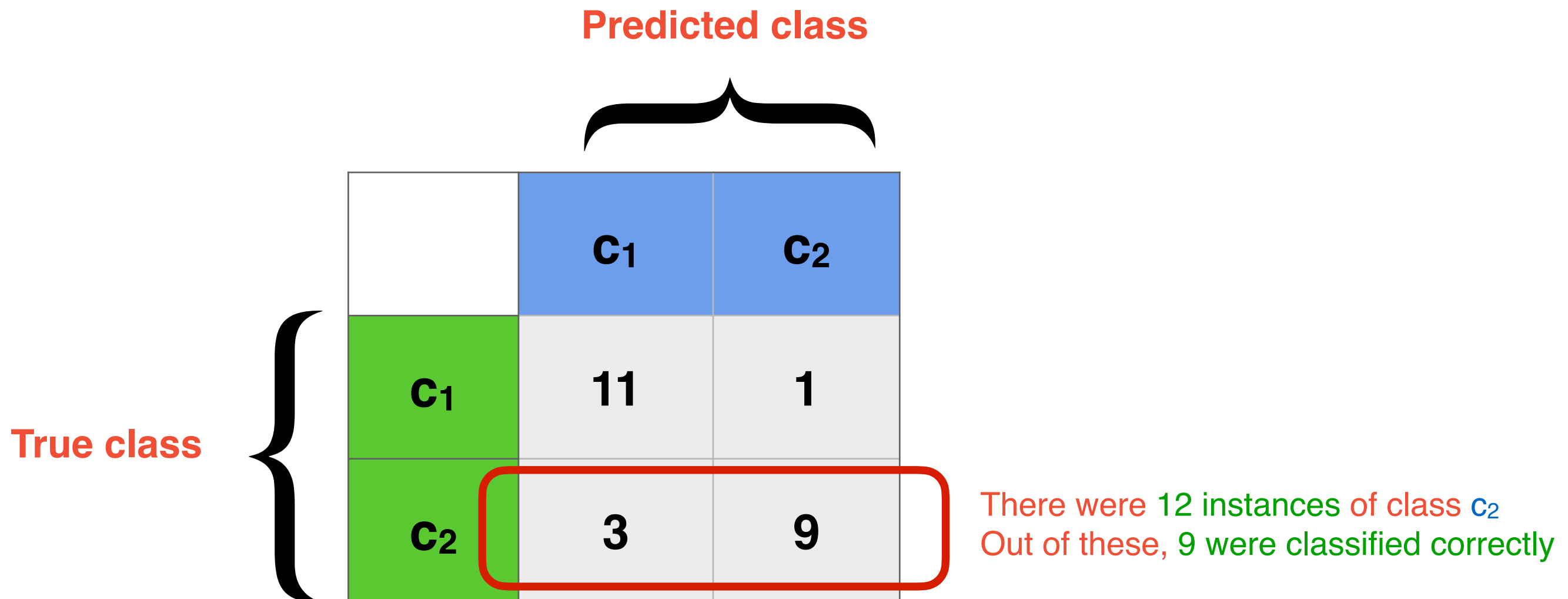
Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)



Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)



Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)
 - Usually there is a “class of interest”, called the **Positive Class (+)**
 - And another class, called the **Negative Class (-)**

True class

Predicted class

	+	-
+	11	1
-	3	9

Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)
 - Usually there is a “class of interest”, called the **Positive Class (+)**
 - And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	11	1
	-	3	9

Quantities of interest:

True Positive (TP)
True Negative (TN)
False Positive (FP)
False Negative (FN)

Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)
 - Usually there is a “class of interest”, called the **Positive Class (+)**
 - And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	11	1
	-	3	9

Quantities of interest:

True Positive (**TP**)
True Negative (**TN**)
False Positive (**FP**)
False Negative (**FN**)

Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)
 - Usually there is a “class of interest”, called the **Positive Class (+)**
 - And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	11	1
	-	3	9

Quantities of interest:

True Positive (TP)
True Negative (TN)
False Positive (FP)
False Negative (FN)

Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
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 - And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	11	1
	-	3	9

Quantities of interest:

True Positive (TP)

True Negative (TN)

False Positive (FP)

False Negative (FN)

Classification Performance

- The performance of a classifier can be visualized as a **confusion matrix**
- Suppose a binary classification problem (two classes only: c_1 and c_2)
 - Usually there is a “class of interest”, called the **Positive Class (+)**
 - And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	11	1
	-	3	9

Quantities of interest:

True Positive (TP)
True Negative (TN)
False Positive (FP)
False Negative (FN)

Confusion Matrix

- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	TP	FN
	-	FP	TN

Quantities of interest:

True Positive (TP)

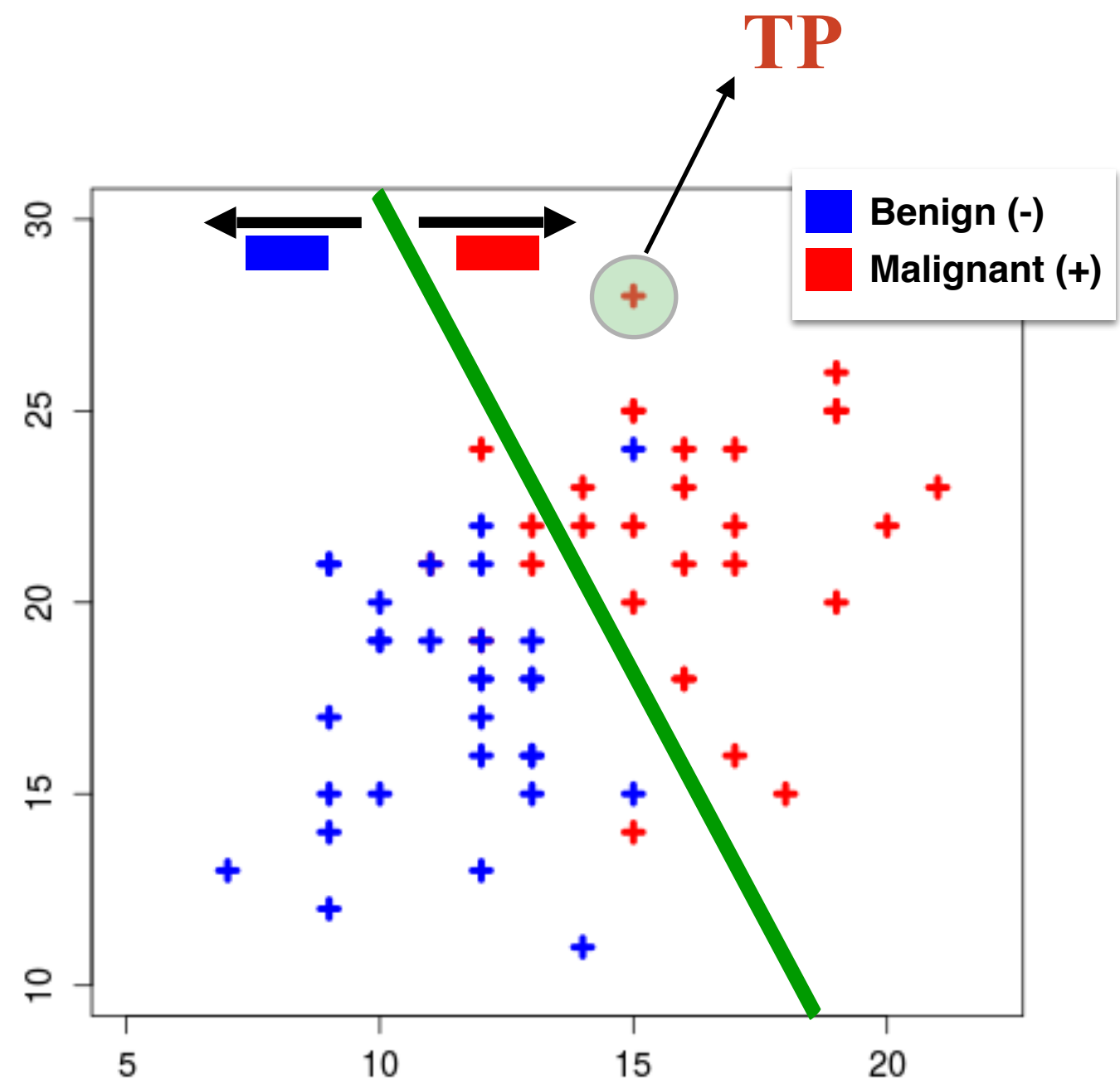
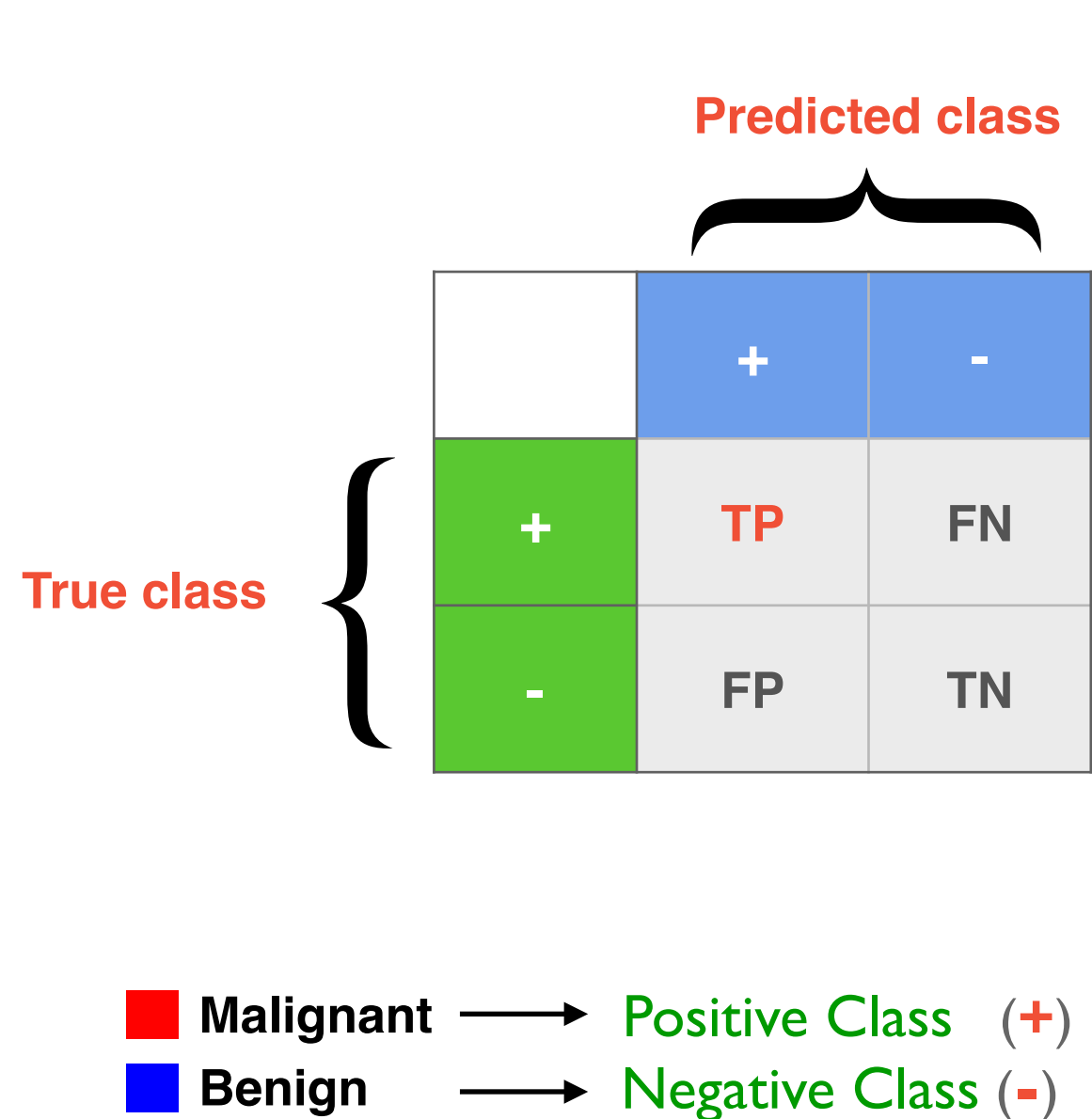
True Negative (TN)

False Positive (FP)

False Negative (FN)

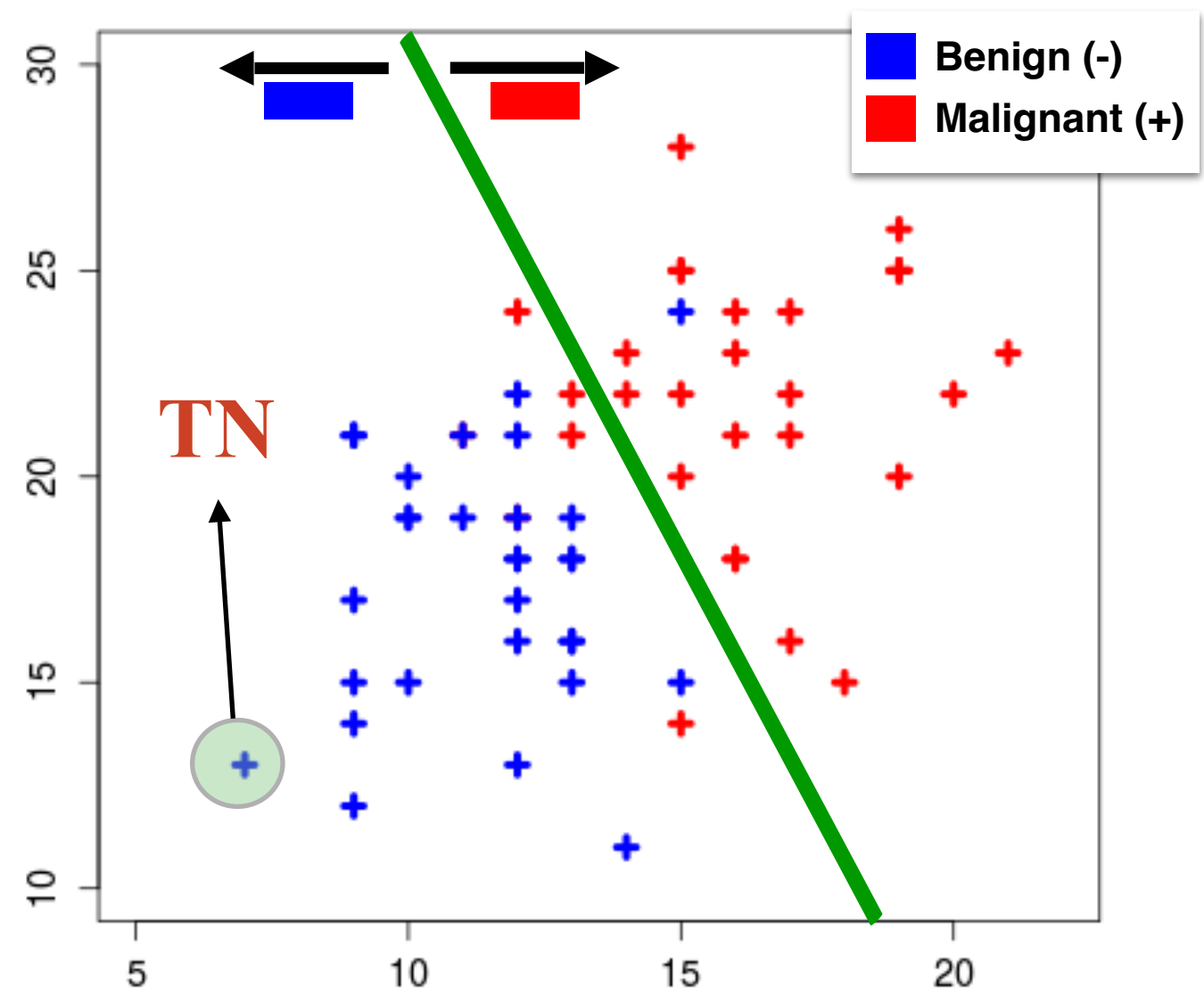
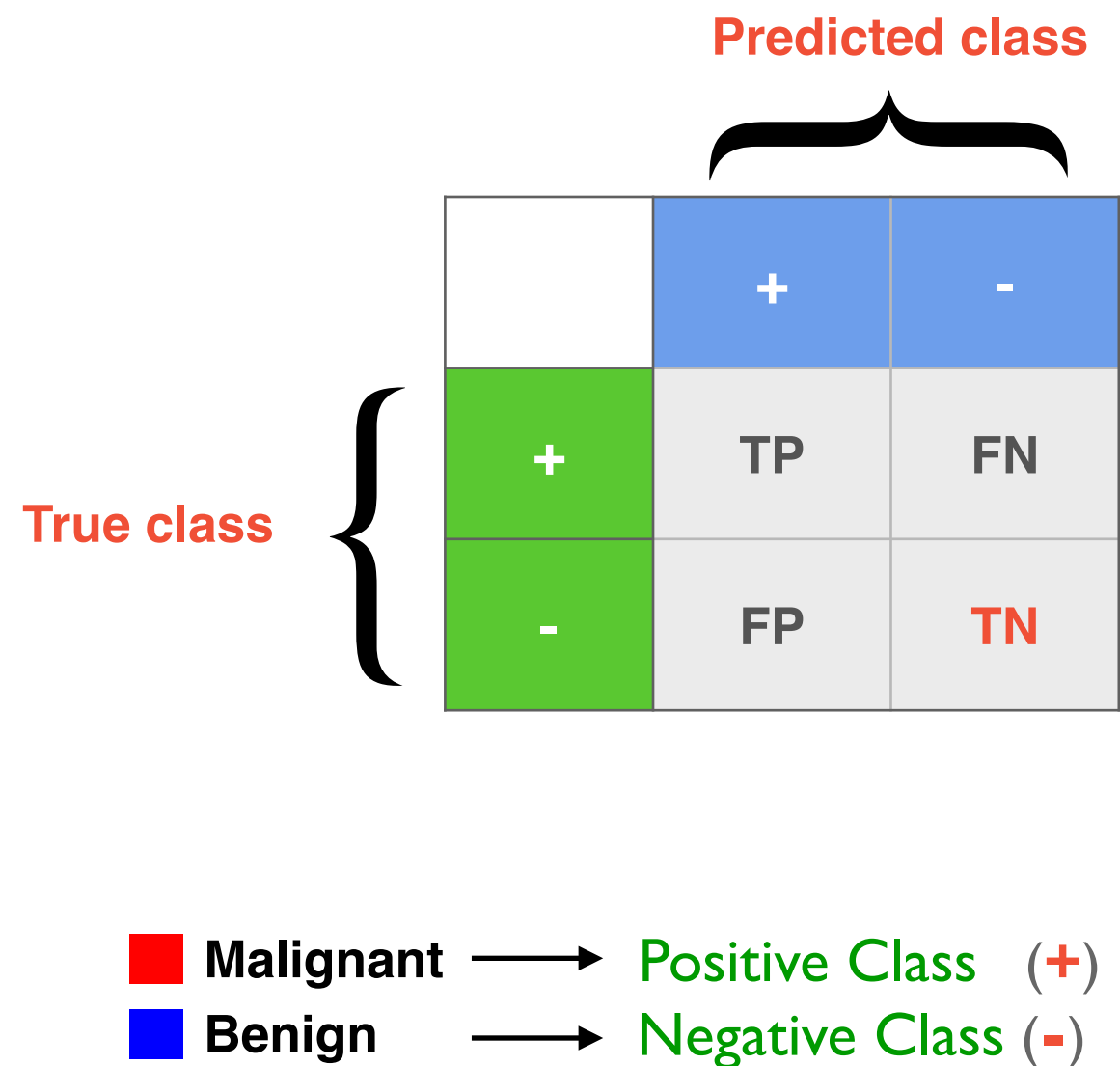
Confusion Matrix

- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**



Confusion Matrix

- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**

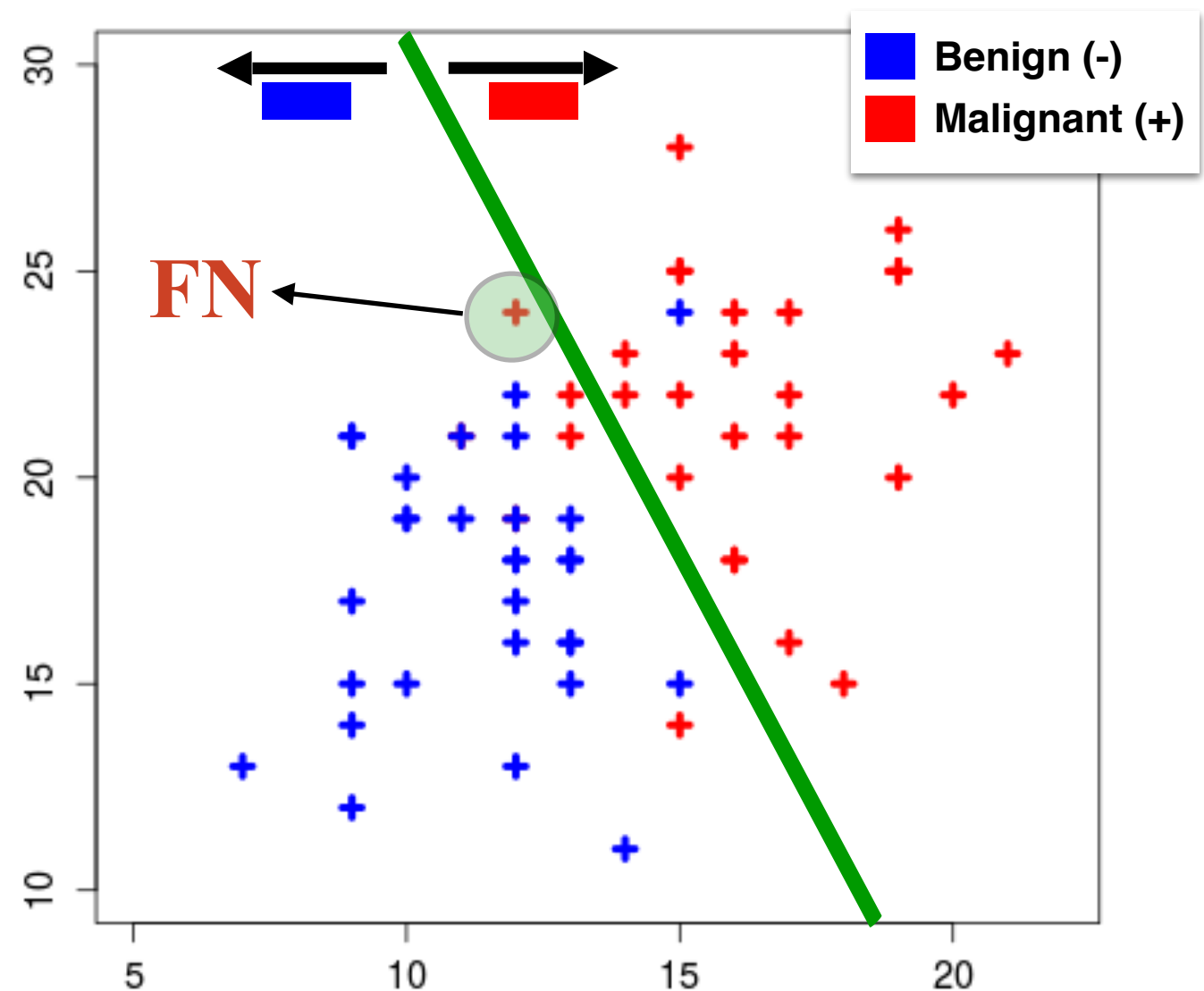


Confusion Matrix

- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**

		Predicted class	
		+	-
True class	+	TP	FN
	-	FP	TN

 Malignant → Positive Class (+)
 Benign → Negative Class (-)



Confusion Matrix

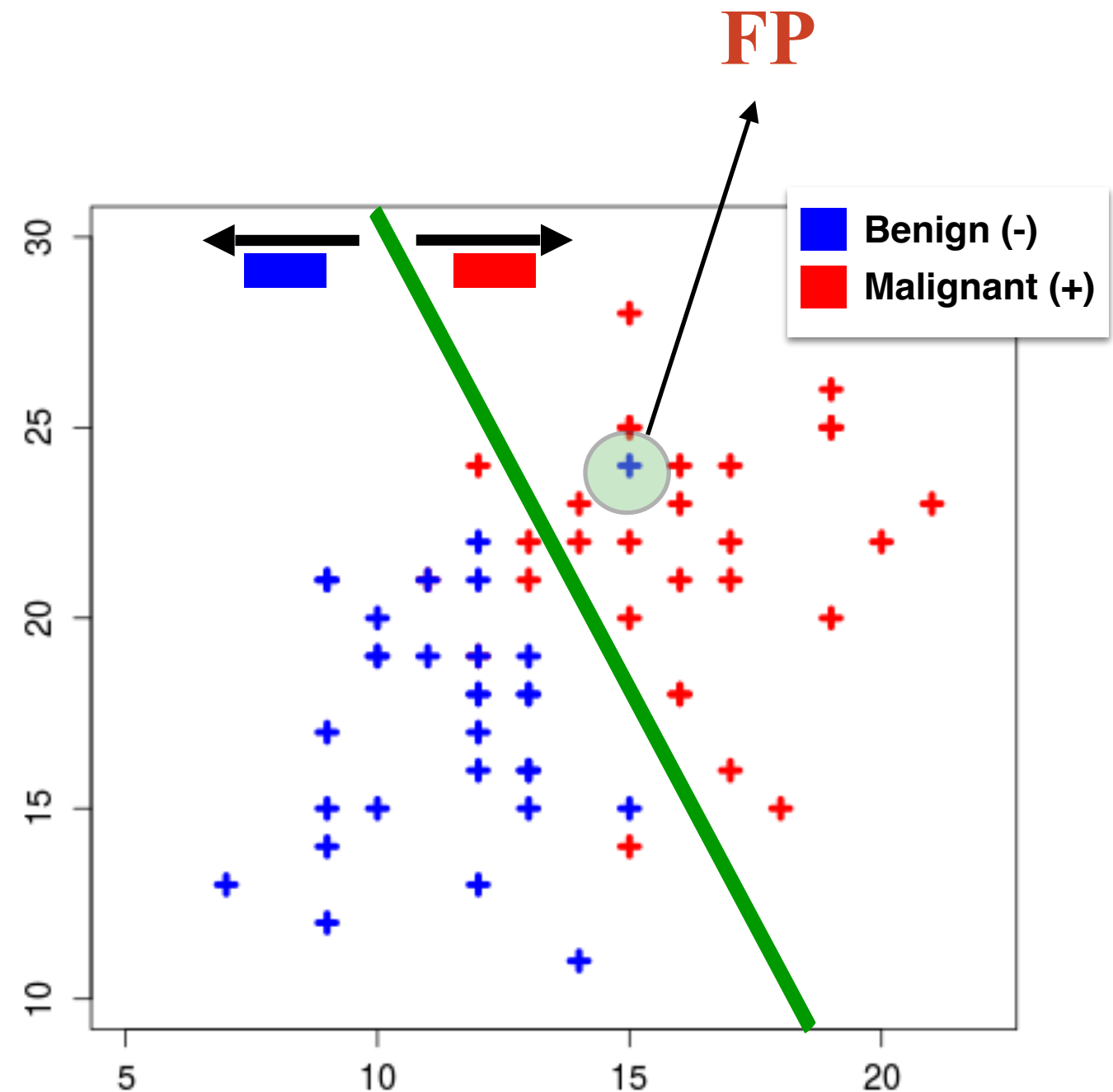
- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**

Predicted class

	+	-
+	TP	FN
-	FP	TN

True class

■ Malignant → Positive Class (+)
■ Benign → Negative Class (-)



Confusion Matrix

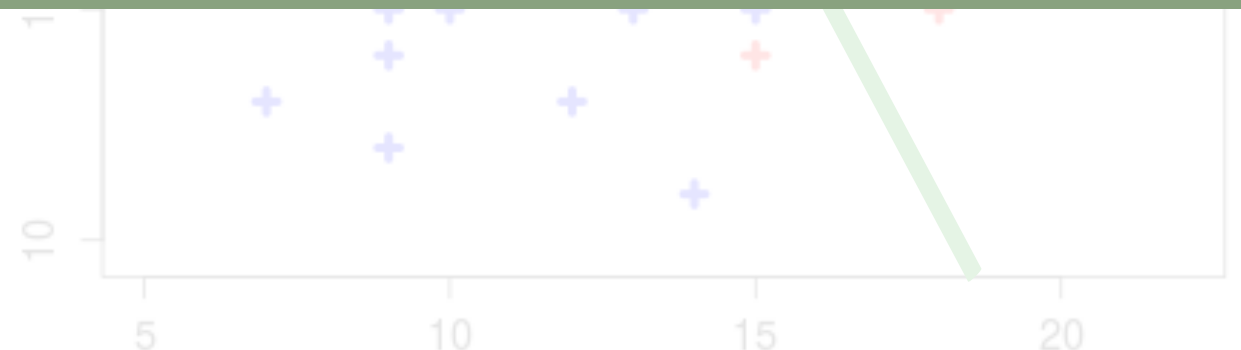
- Usually there is a “class of interest”, called the **Positive Class (+)**
- And another class, called the **Negative Class (-)**

Predicted class



- **True Positives** (TP): positive instances correctly classified as positive
- **True Negatives** (TN): negative instances correctly classified as negative
- **False Positives** (FP): negative instances incorrectly classified as positive
- **False Negatives** (FN): positive instances incorrectly classified as negative

■ Malignant → Positive Class (+)
■ Benign → Negative Class (-)





Machine Learning

CMPSCI 589

Bruno C. da Silva

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Evaluation Metrics (1/2)

UMassAmherst
College of Information
& Computer Sciences



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